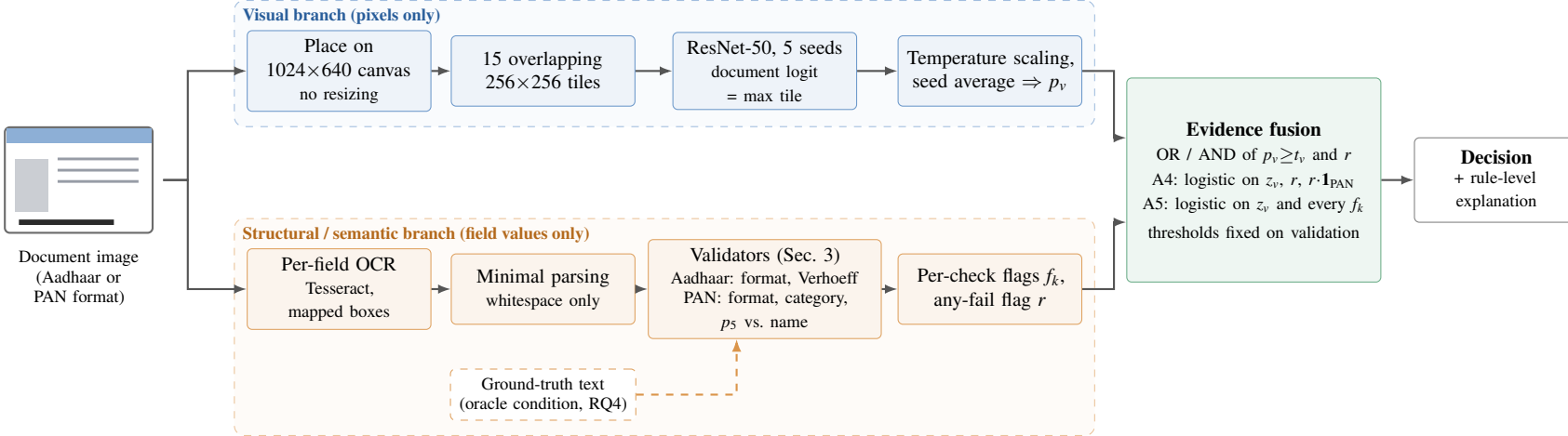
Visual branch (pixels only)
Place on 1024×640 canvas no resizing
15 overlapping 256×256 tiles
ResNet-50, 5 seeds document logit = max tile
Temperature scaling, seed average ⇒ p_v
Document image (Aadhaar or PAN format)
Structural / semantic branch (field values only)
Per-field OCR Tesseract, mapped boxes
Minimal parsing whitespace only
Validators (Sec. 3) Aadhaar: format, Verhoeff PAN: format, category, p_5 vs. name
Per-check flags f_k, any-fail flag r
Ground-truth text (oracle condition, RQ4)
Evidence fusion
OR / AND of $p_v \geq t_v$ and r
A4: logistic on z_v, r, $r \cdot \mathbf{1}_{\text{PAN}}$
A5: logistic on z_v and every f_k
thresholds fixed on validation
Decision
+ rule-level explanation